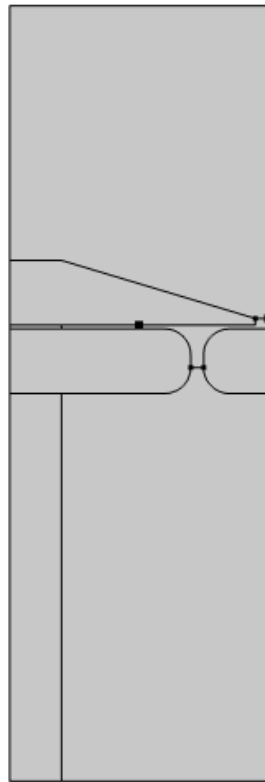


Simulation Exercise 3 - Bicycle damper

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Abstract

The study investigates the fluid–structure interaction mechanisms that govern the suspension damper of a mountain bike. Although the assignment describes a front-fork damper, the modeled system corresponds to a real rear damper. Viscous fluid dampers control wheel motion by forcing oil through ports and flexible shim stacks, which deform under pressure and regulate the damping force.

In the first part of the project, the shims are assumed rigid in order to isolate and study the pressure distribution and flow behavior inside the cylinder without structural deformation. A 2D axisymmetric FEM model is created to capture the fluid motion, pressure gradients, and resulting force–velocity relation.

In the second part, the analysis is extended by modeling the shims as deformable solid components made of high-strength steel. A fully coupled fluid–structure interaction (FSI) model is used, allowing the oil domain to deform with the structure. The influence of shim flexibility, shim count, and geometry on the force–velocity response is studied, and the damping force is fitted to $F = C|v_{in}/v_0|^m$. Changes in m and C with different parameter values are investigated.

Finally, a full 3D model is developed to capture effects that cannot be represented in axisymmetric simulations. Since real shim stacks block reversed flow asymmetrically, the damper is not axisymmetric in practice. A 3D representation of the port geometry is therefore created to resolve circumferential flow variations and partial flow blockage. The resulting 3D force–velocity curves are compared with the 2D axisymmetric predictions from the earlier cases.

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1. Problem background and definition

A bicycle damper aims to absorb sudden shocks that may arise while riding a bike in uneven terrain. This study aims to investigate the fluid-structure interaction inside a damper to investigate how the geometry of the damper, number of shims et cetera impact the resultant force-velocity curves.

2. Model definition

2.1 Geometry and materials

The geometry of the structure is shown in Figure 1, while the parameters used are presented in Table 1. The chamber is filled with oil, while other parts are made of steel. When a parameter is presented in a range the largest value is selected.

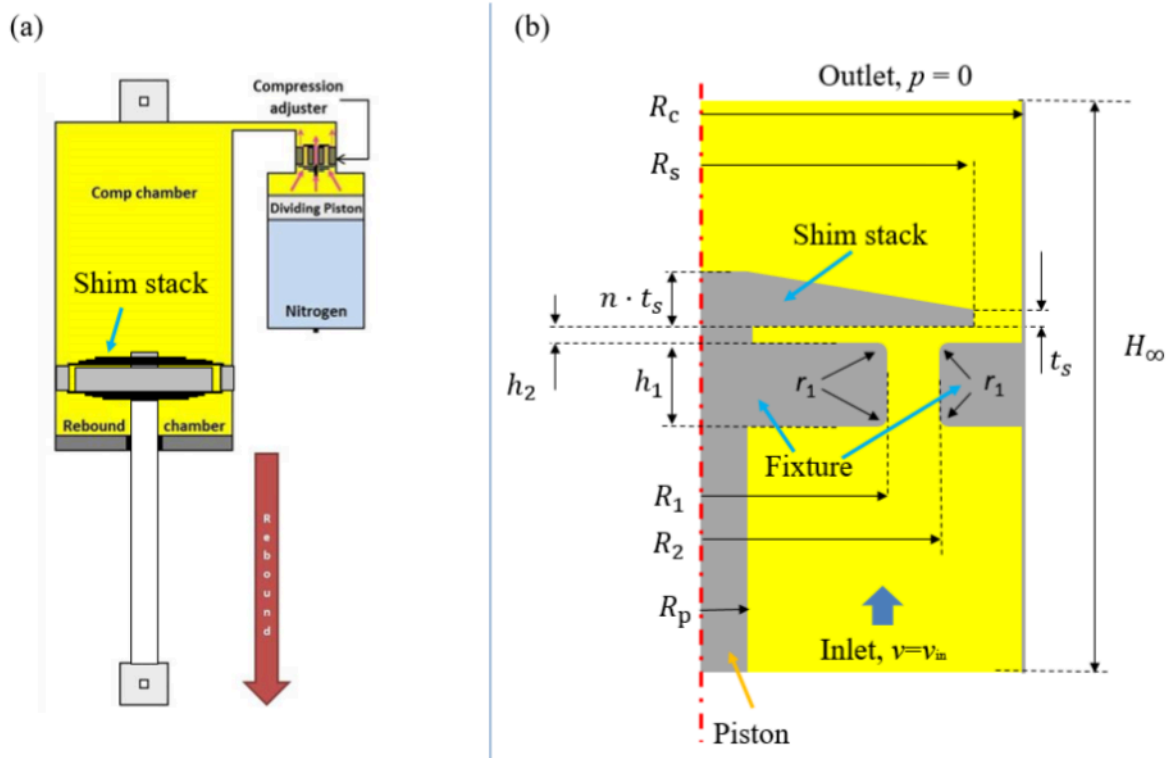


Figure 1: Dimensions and geometry of the damper. Source: Project assignment page on Canvas.

Table 1: Parameters used for the assignment

Parameter	Value		Comment
R_c	20	mm	Inner radius of damper
R_p	4	mm	Piston Radius
R_s	15-19	mm	Radius of largest shim
R_l	9-14	mm	
R_2	15	mm	
t_s	0.2-0.5	mm	Thickness of one shim
n	1-10		Number of shims
r_l	0.5-0.2	mm	Fillet radius
h_l	5	mm	Height of structure
h_{2a}	0.2-0.3	mm	
h_{2b}	0.05-0.3	mm	
h_{2c}	0.05-0.3	mm	
H_{inf}	60	mm	Total simulation of height
μ (μ)	0.1	Ns/m ²	Shear viscosity of the oil
ρ (ρ)	850	kg/m ³	Density of the oil
E	200	GPa	Young's Modulus
ν (ν)	0.3		Poisson's ratio
v_{in}	0-1	m/s	Global speed of the damper

2.2.1: Fluid mechanics equations

The simulation assumes an incompressible Newtonian fluid under laminar flow as turbulent flow is complex and demanding. Laminar flow occurs when the Reynolds number is sufficiently low, which is under around $Re = 1000$. The formula for Reynolds number is featured in Eq. 1.

$$R_e = \frac{\rho UL}{\mu} \quad (1)$$

Where U is the velocity, L is the characteristic length, ρ is the density and μ is the viscosity of the fluid. The main equations of fluids that are considered in this simulation exercise are rate of deformation, velocity equations, conservation of mass or continuity equation, equation of motion and constitutive equations. These equations together form the Navier-Stokes equation (eq. 2).

$$\rho \left(\frac{\partial v_i}{\partial t} + v_j \frac{\partial v_i}{\partial x_j} \right) = \mu \left(\frac{\partial^2 v_i}{\partial x_j^2} + \frac{1}{3} \frac{\partial}{\partial x_i} \left(\frac{\partial v_j}{\partial x_j} \right) \right) - \frac{\partial p}{\partial x_i} + \rho f_i \quad (2)$$

Where ρ is the density of the fluid (kg/m^3), μ is the dynamic viscosity ($\text{Pa}\cdot\text{s}$), p is the pressure field (Pa), $\mathbf{v} = (v_x, v_y, v_z)$ is the velocity vector of the fluid (m/s), v_i is the velocity component in direction x_i (Einstein notation), x_i is the spatial coordinate (x, y, z), t is the time (s), f are external forces per unit mass (e.g., gravity), and f_i is the component of external force (m/s^2). The Navier-Stokes equation is difficult to solve without a finite element model (FEM) program. The governing continuity equation for fluid flow in isothermal conditions, the Euler spatial model.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0; \quad \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x_k} (\rho v_k) = 0 \quad (3)$$

Where variables are defined the same as above, as in Eq. 2.

2.2.2: Fluid structure interaction (FSI):

Fluid–structure interaction (FSI), present in cases (b) and (c), involves the coupling between a deforming or moving structure and surrounding fluid. The FSI domain is split into fluid and solid domains that share a common interface. Fluid motion is governed by the Navier–Stokes and continuity equations (Section 2.2.1), with the Navier–Stokes equations describing the evolution of fluid velocity and the fluid stress tensor given by Eq. 4.

$$\boldsymbol{\sigma}_F = \mu \left(\nabla \mathbf{v}_F + (\nabla \mathbf{v}_F)^T \right) - p_F \mathbf{I} \quad (4)$$

The solid is assumed, in this simulation exercise to be elastic, the solid structure can thus be represented by Eq. 5.

$$\rho_s \frac{\partial \mathbf{v}_s}{\partial t} = \nabla \cdot \boldsymbol{\sigma}_s + \rho \mathbf{f}_s \quad (5)$$

Where the stress tensor is derived from strain-energy density function $\Psi(b)$, for an isotropic cauchy elastic material.

$$\boldsymbol{\sigma}_s = \Psi(\mathbf{b}); \mathbf{b} = \mathbf{F}\mathbf{F}^T - \mathbf{I}; \mathbf{F} = \frac{\partial(\mathbf{X} + \mathbf{u})}{\partial \mathbf{X}} \quad (6)$$

The fluid–solid interface boundary is coupled through dynamic and kinematic constraints, i.e. traction continuity (Eq. 7) and velocity continuity (Eq. 8), which must be satisfied at the boundary.

$$t_s = -t_f = \boldsymbol{\sigma}_F \cdot \mathbf{n} \quad (7)$$

$$v_F = v_s \quad (8)$$

The definitions of the FSI boundaries are found in Figure 2. For a two-way coupling, the motion of the solid affects the fluid (changing the fluid mesh, pressure, and velocity), and the forces from the fluid affect the deformation of the solid.

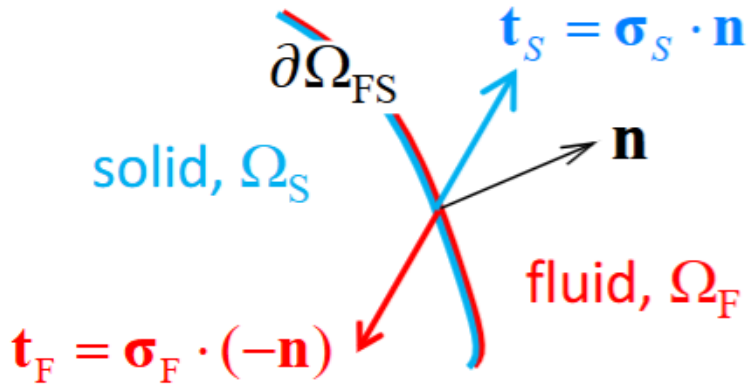


Figure 2: Fluid and solid boundaries of the two-way FSI coupling.

In COMSOL, mesh motion is handled using the Arbitrary Lagrangian–Eulerian (ALE) framework, a moving mesh module. The structure is treated with a Lagrangian description and the fluid with an Eulerian one, allowing the mesh to smoothly accommodate interface motion. A purely Lagrangian approach risks severe mesh distortion during large deformations, but combining it with the Eulerian description adjusts node positions at the moving interface and preserves mesh quality. A moving mesh is essential in FSI because a fixed mesh cannot follow structural motion, causing simulation failure or requiring constant remeshing.

2.3 Boundary conditions

Case (a): only the fluid is modeled, solid mechanics and FSI multiphysics are disabled, and a 2D axisymmetric setup is used. Laminar flow is applied, with fixed walls on the shimstack and fixtures. The right-hand “outer wall” is defined as a sliding wall with velocity $-v_{in}$ in the defined direction (Fig. 3). The inlet (lower edge) is given an initial velocity v_{in} , and the outlet (upper boundary) is set to zero pressure. See 3 for full boundary conditions.

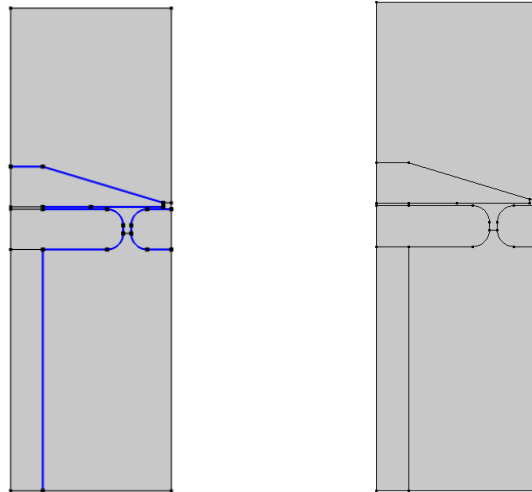


Figure 3: Boundary conditions marked in blue for case (a). Left: fixed wall constraint. Right: sliding wall constraint with velocity $v = -v_{in}$.

Case (b): The same boundary conditions as in case (a) were used, but a fixed constraint was added in the solid mechanics module (see Figure 4). Solid mechanics was applied to all parts not included in the laminar flow module.

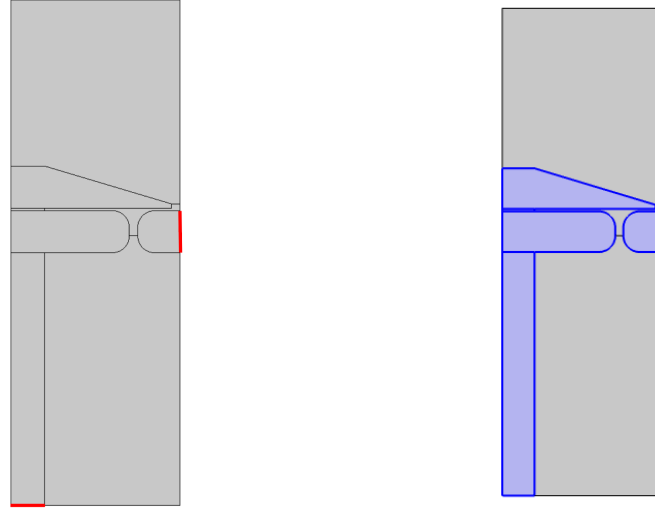


Figure 4: Boundary conditions of structure. Left: fixed boundary condition in red. Right: all parts to which solid mechanics (solid) was applied

Case (c): A 3D model was used in COMSOL which allowed for modeling of the real geometry with non-axisymmetry. Symmetry boundary conditions were used so that only one fourth of the geometry was needed as seen in Figure 5. Otherwise, the same boundary conditions as in (b) were used for the 3D model, the conditions were applied to the boundaries rather than edges.

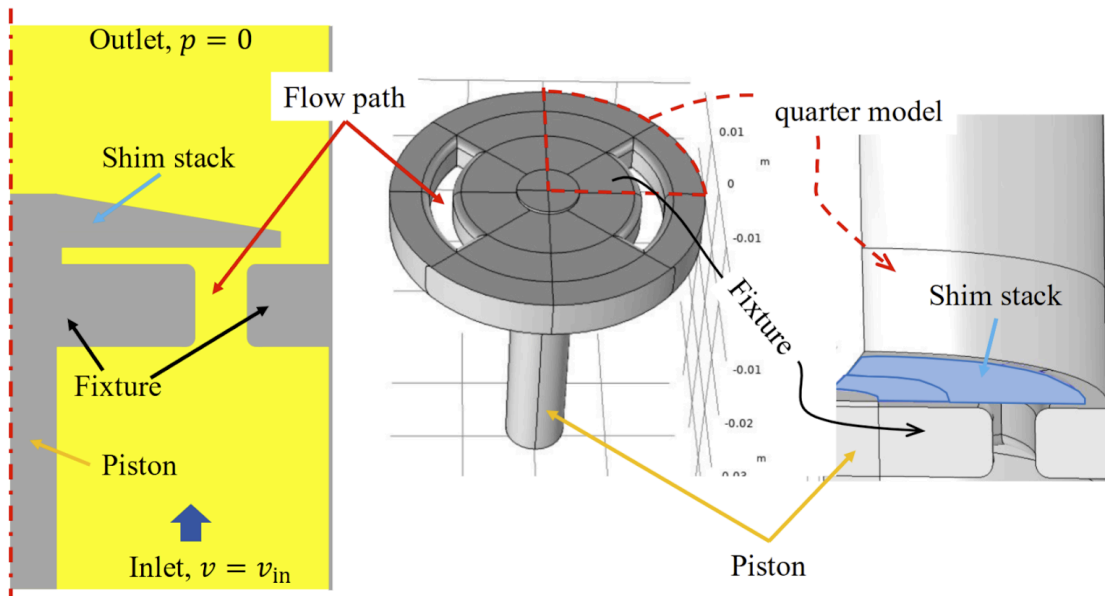


Figure 5: Geometry used for case c

3. Numerical model

The numerical model was created using values from Table 1. Free triangular mesh was applied to the upper and lower parts, while a second finer free triangular mesh was applied to the middle section where the highest velocity was expected. Lastly a boundary layer mesh was applied to the inside right side wall. See Figure 6.

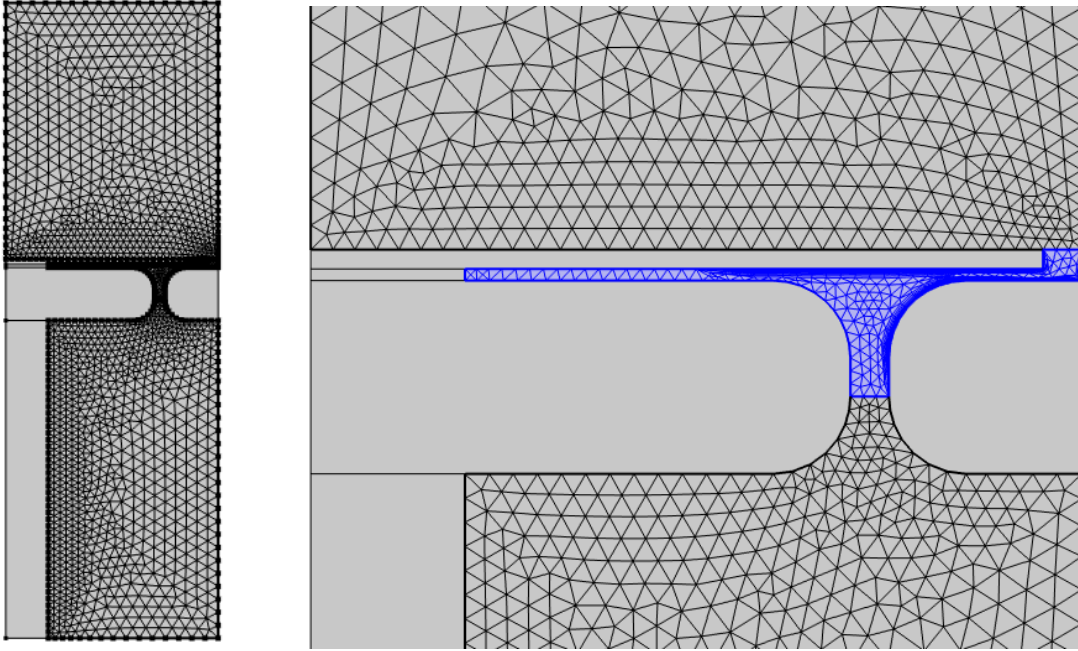


Figure 6: *Left: Full mesh used during the simulations. Right: second finer free triangular elements.*

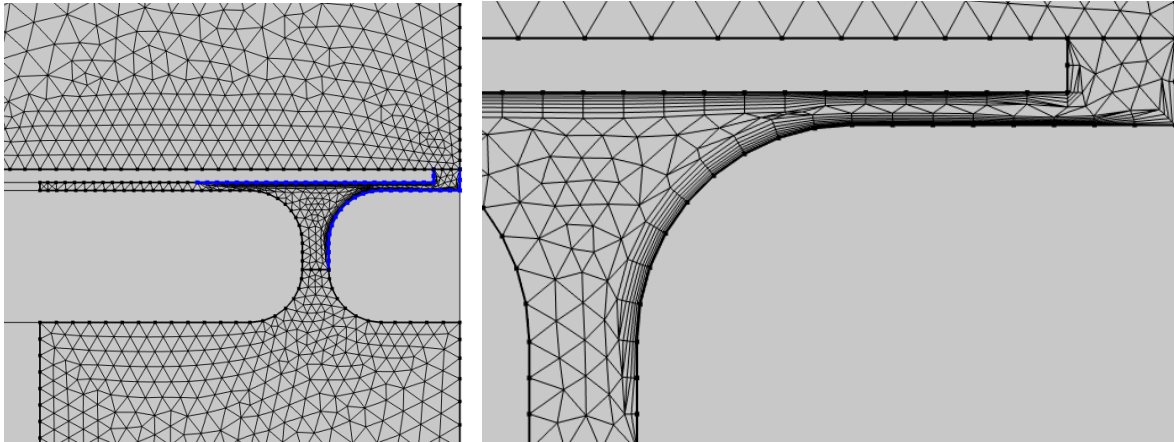


Figure 7: *Left: The boundaries that boundary layer meshing was applied to. Right: Boundary layers meshing.*

The number of boundary layers was controlled by the parameter *numMeshLayers* (=8 in Figure 7). The maximum element sizes were scaled as $0.005/\text{numMeshLayers}$ for the coarse region and $0.003/\text{numMeshLayers}$ for the finer middle section, other meshing parameters were set to COMSOL prescribed. Mesh convergence was assessed through a parametric sweep (Chapter 4).

Case (b): Moving mesh module is added to the fluid to allow for deformation of the shimstack. The mesh in the fluid region was set to the same as for case (a). The shimstack domain was set a maximum size $0.003/\text{numMeshLayers}$ and the rest of the solid was set a max size 0.01. Mesh is seen in Appendix [A].

Case (c): The mesh used for case (c) is similar to case (b) but with changed values. The parameter numMeshLayers is still the number of boundary layers. The full fluid region and shimstack used a max element size $0.008/\text{numMeshLayers}$ while the rest of the solid used max size 0.01 m since this domain will not deform.

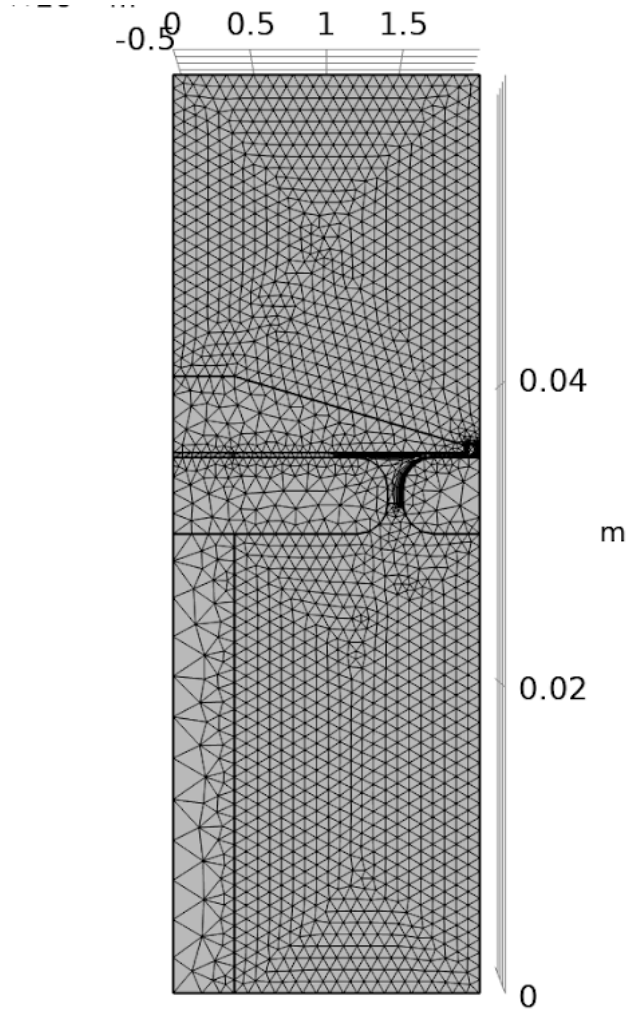


Figure 8: Mesh used for case c

4. Results and Discussion

4.1. Results for case (a)

4.1.1: Analytical solution for case (a)

An analytical solution assuming constant average volume flow was used to determine the average velocity. In COMSOL, the average velocity was obtained via a 1D plot group and the average function over the smallest region.

$$Q = A * v \quad (9)$$

For an incompressible fluid, the volume flow Q is constant and thus one can compare the inlet and the thinnest regions.

$$Q = A_{inlet} * v_{inlet} = A_{thin} * v_{thin} \quad (10)$$

The formula was implemented in MATLAB (see appendix), yielding an average velocity of $v = 37.7$ m/s for the thinnest section, compared to $v = 37.37$ m/s in COMSOL. This shows that, for the laminar flow case, the analytical solution provides a good approximation. The average velocity in the thinnest region is below 60 m/s, as required by the assignment.

4.1.2: Mesh convergence

Figure 9 shows that mesh convergence improves rapidly as the number of boundary layers increases. After $numMeshLayers = 8$, it varies by about 100 N (~3%), indicating convergence thus this mesh was used both for case (a) and (b).

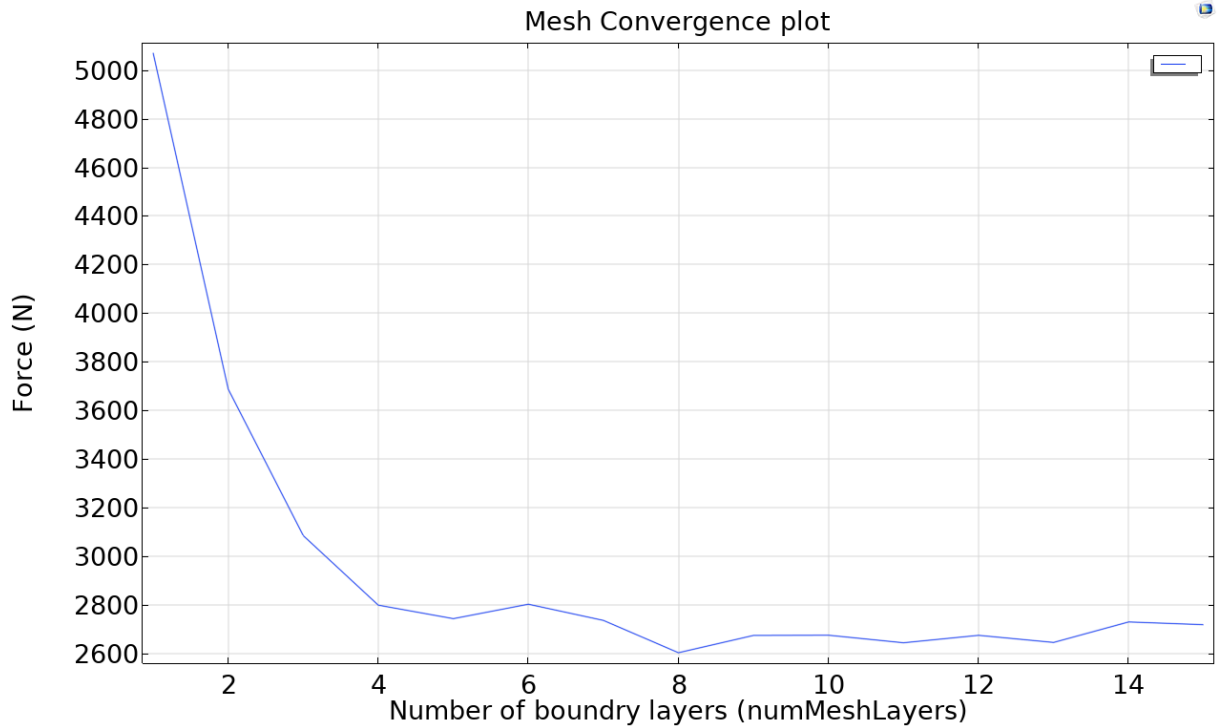


Figure 9: Mesh convergence study for $v_{in} = 1$ m/s, all other parameters kept constant.

Convergence depends strongly on the parameters. For example, with an initial velocity of 0.1 m/s, the mesh can be considered converged immediately, as the largest changes are only

2–3%, even though the convergence plot appears poor (Figure 10), note that the plot only changes from 175 to 177.6 Newton.

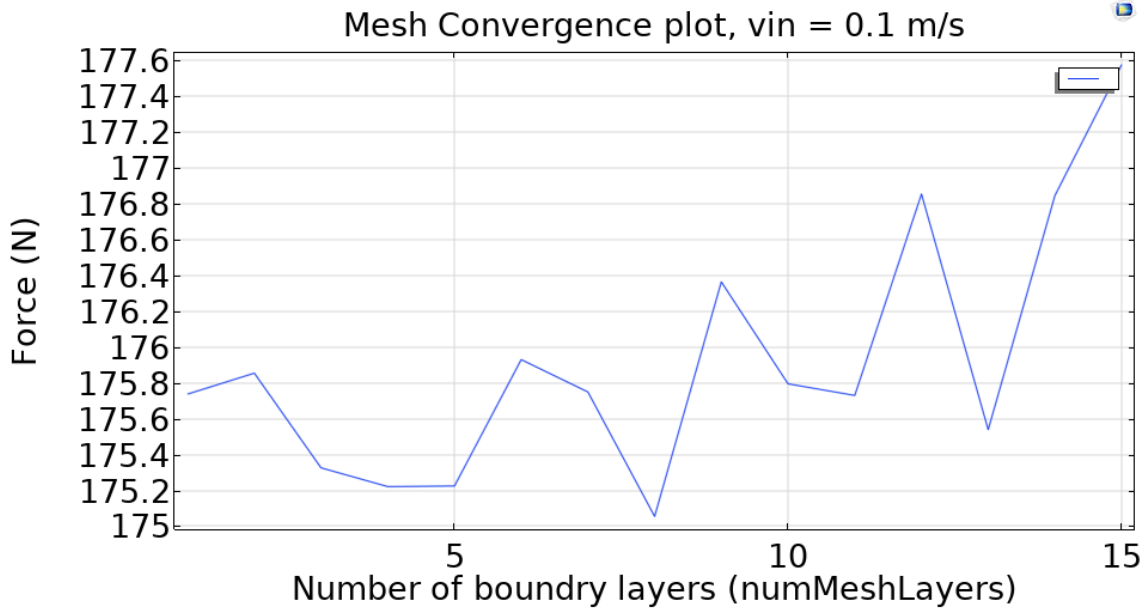


Figure 10: Mesh convergence study for $v = 0.1$ m/s, all other parameters kept constant.

4.1.3: Force-velocity plots for case (a)

Equilibrium over the whole fluid domain in z-direction is $F = \int_{A_{in}} p_{in} dA - \int_{A_{out}} p_{in} dA$ with A

and p being area and pressure of the inlet and outlet respectively and F being the force from the piston on the fluid in negative z-direction. With $p_{out} = 0$ the force is calculated by integrating the pressure over the inlet.

In case (a), the height between the shimstack and fixtures (h_{2a}), see Figure 1, was varied.

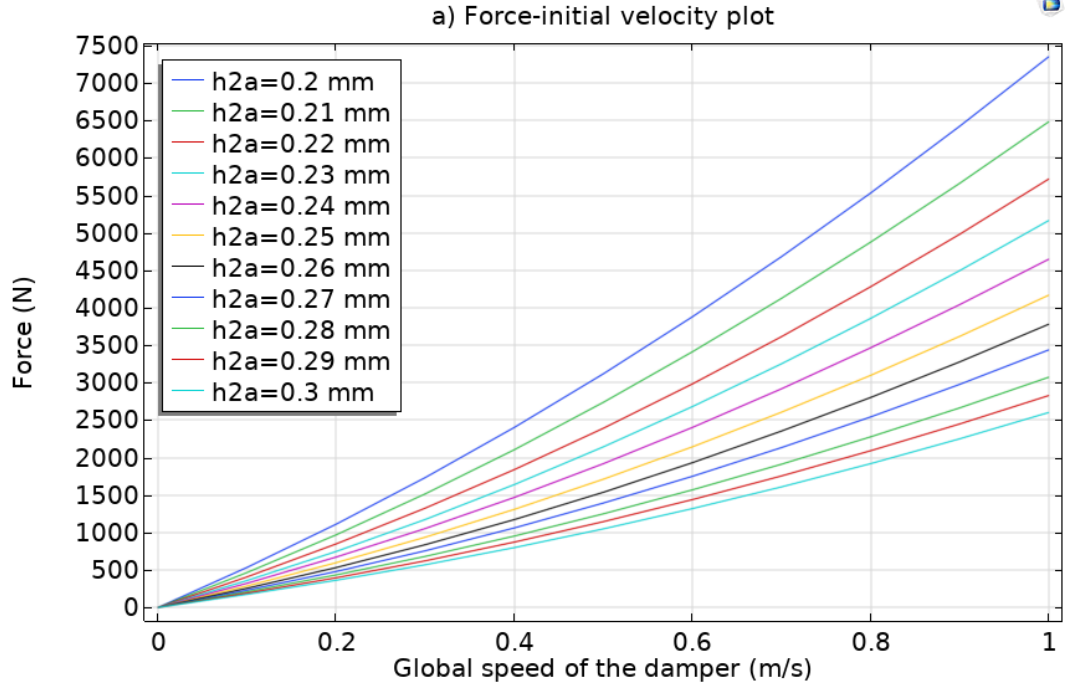


Figure 11: Force-velocity graphs when h_{2a} was varied.

Figure 11 shows that increasing h_{2a} decreases the force. A smaller h_{2a} (0.2 mm) creates a larger pressure gradient in the shimstack, increasing local velocity and, consequently, the surface forces as seen in Figure 12.

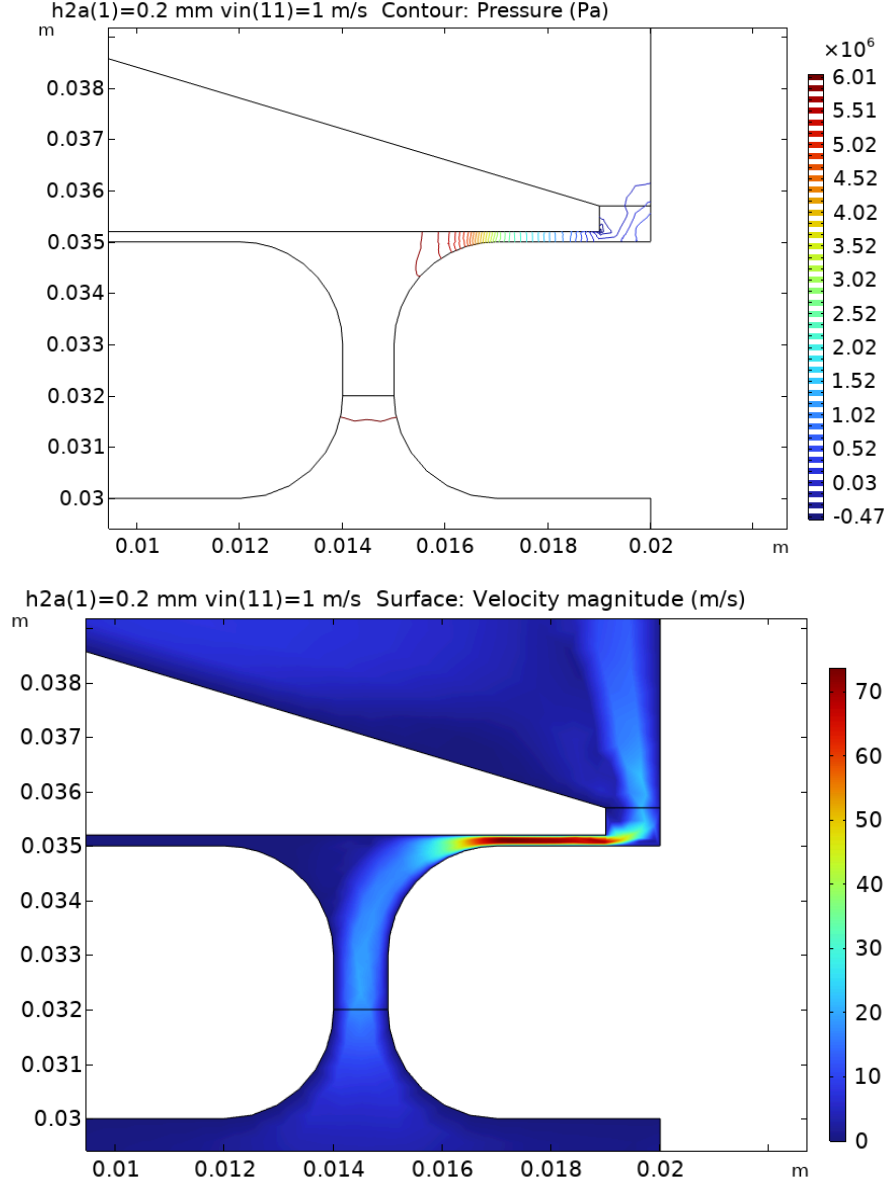


Figure 12: Upper: Contour pressure plot. Lower: Velocity magnitude plot, both plots are for $h2a = 0.2 \text{ mm}$ and $v_{in} = 1 \text{ m/s}$.

The data for force-velocity plots (Figure 11) for different $h2a$ were fitted to the expression $F = C|v_{in}/v_0|^m$, using “lsqcurvefit” in matlab and yielded the values for m and C in Table 2.

Table 2: Force velocity with varying height between shimstack and piston fixture for case (a).

$h2a$ [mm]	m [-]	C [N]
0.20	1.209	7288
0.21	1.216	6430
0.22	1.226	5664
0.23	1.241	5119
0.24	1.245	4605
0.25	1.252	4126
0.26	1.265	3743
0.27	1.271	3402
0.28	1.265	3041
0.29	1.271	2799
0.30	1.277	2574

The expression for F was also plotted in matlab for the different $h2a$ and resulted in similar plots as in Fig 11 validating the results for m and C . See Figure 13.

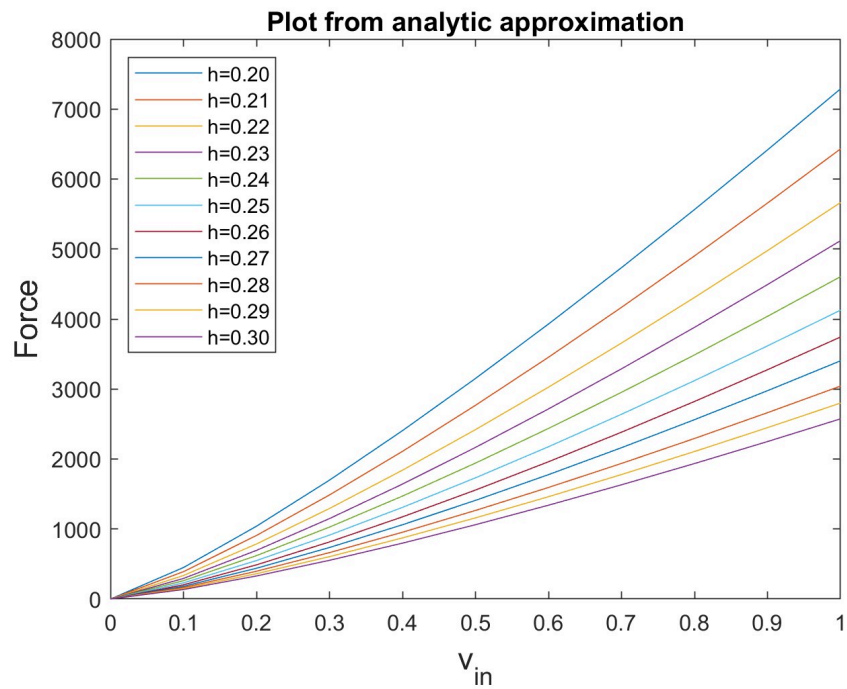


Figure 13: curve-fitting plots for different $h2a$ values.

The C represents our maximum force, when $v_{in} = v_0 = 1 \text{ m/s}$. The m is 1.2-1.3 for all $h2a$, a m close to one means that the pressure difference is viscosity dominant while a m close to 2 means that the pressure difference is largely dependent on the inertia.

The relation between force and velocity was also investigated for different numbers of shims and yielded the results in Figure 14 with m and C in Table 2. Since this case has rigid shims the number of shims change the results little.

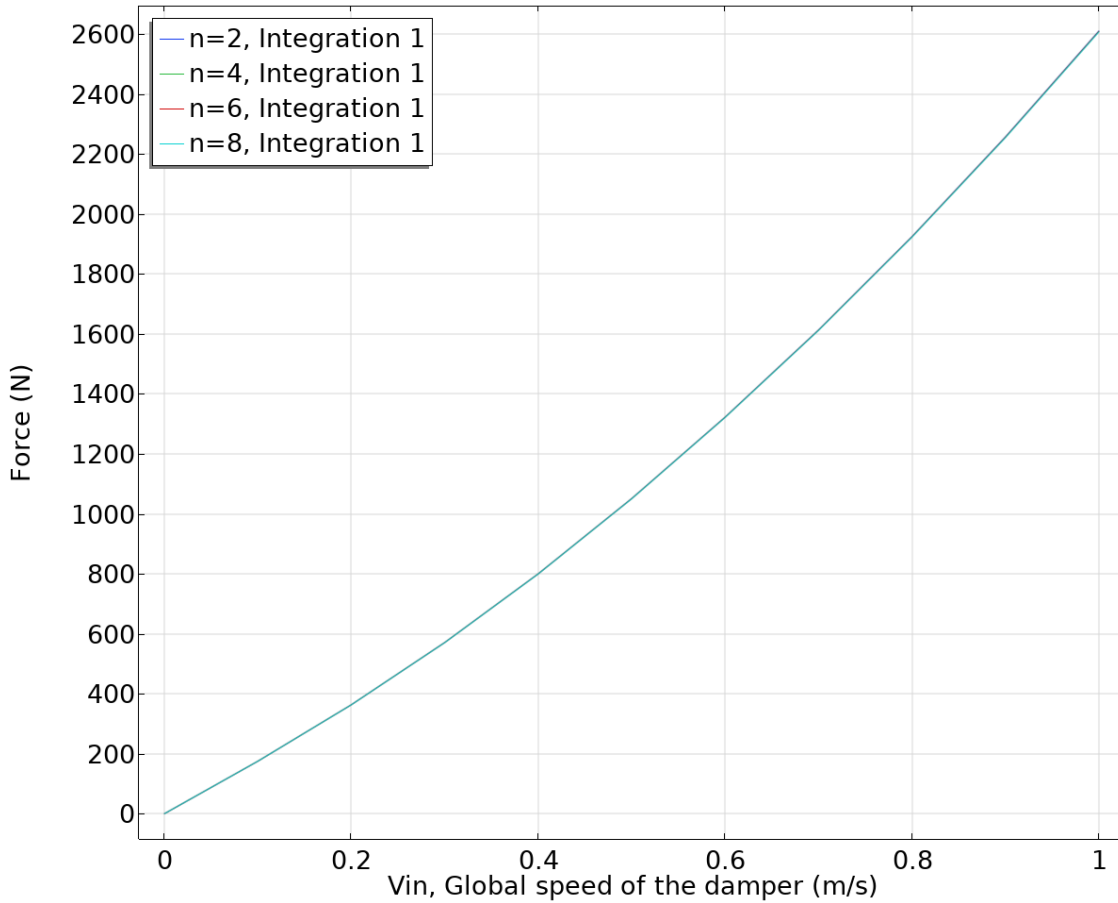


Figure 14: Force-velocity plots for varying n

4.1.3: Creeping flow

Convective acceleration was excluded from the Navier-Stokes equation by omitting inertia terms in COMSOL's laminar flow module. The results are shown in Figure 15.

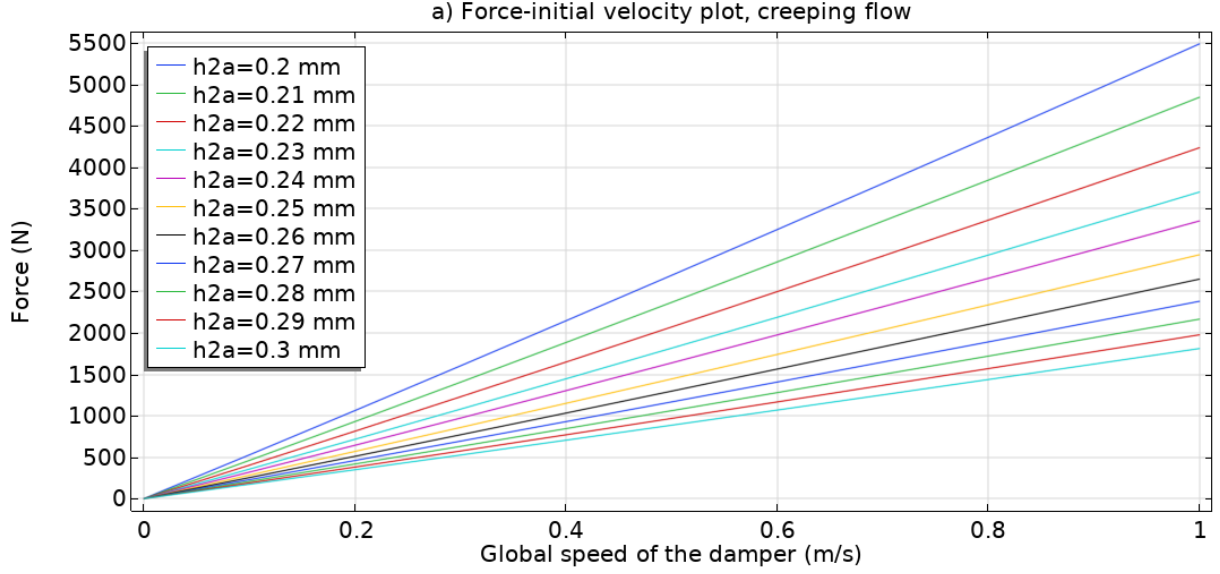


Figure 15: Varying $h2a$, inertial terms in Navier-Stokes excluded.

From Figure 15, it is seen that the plots are linear ($m=1$), compared to the plots featured in Figure 11, comparing the two figures one can observe that the force amplitudes are significantly lower when interior terms are excluded, for example $h2a = 0.2$ mm and $v_{in} = 1$ m/s, gives a value of 5500 N in Figure 15, while it gives a value of nearly 7500 N in Figure 11. The assumption of creeping flow requires a Reynolds number much lower than 1, which is not the case in the simulation due to high velocity and small gaps compared to inlet, which in turn expects a Reynolds number larger than 1. The inertial term also scales with v^2 rather than v .

4.2. Result for case b

Investigation of the changes for C and m with a flexible shimstack were carried out with varying the height of the gap in the same manner as in case a. This gave the results in Table 3.

Table 3: Force velocity with varying height between shim stack and piston fixture for case (b).

$h2a$ [mm]	m [-]	C [N]
0.20	1.114	6306
0.21	1.148	7006
0.22	1.172	5837
0.23	1.137	4467
0.24	1.127	4556
0.25	1.122	4027
0.26	1.189	3421
0.27	1.151	3156
0.28	1.186	2848
0.29	1.138	2892
0.30	1.172	2527

Comparing Table 3 with Table 2, shows that allowing deformation of the shimstack results in a lower force, C , and a slightly lower exponent, m . This is because the deformation of the shimstack allows for a greater area for the flow which decreases the resistance.

In case (b), Figure 16 shows that increasing the number of shims n increases the force. Fewer shims allow greater deformation, increasing the gap ($h2a$) and thus reducing the force once steady state is reached.

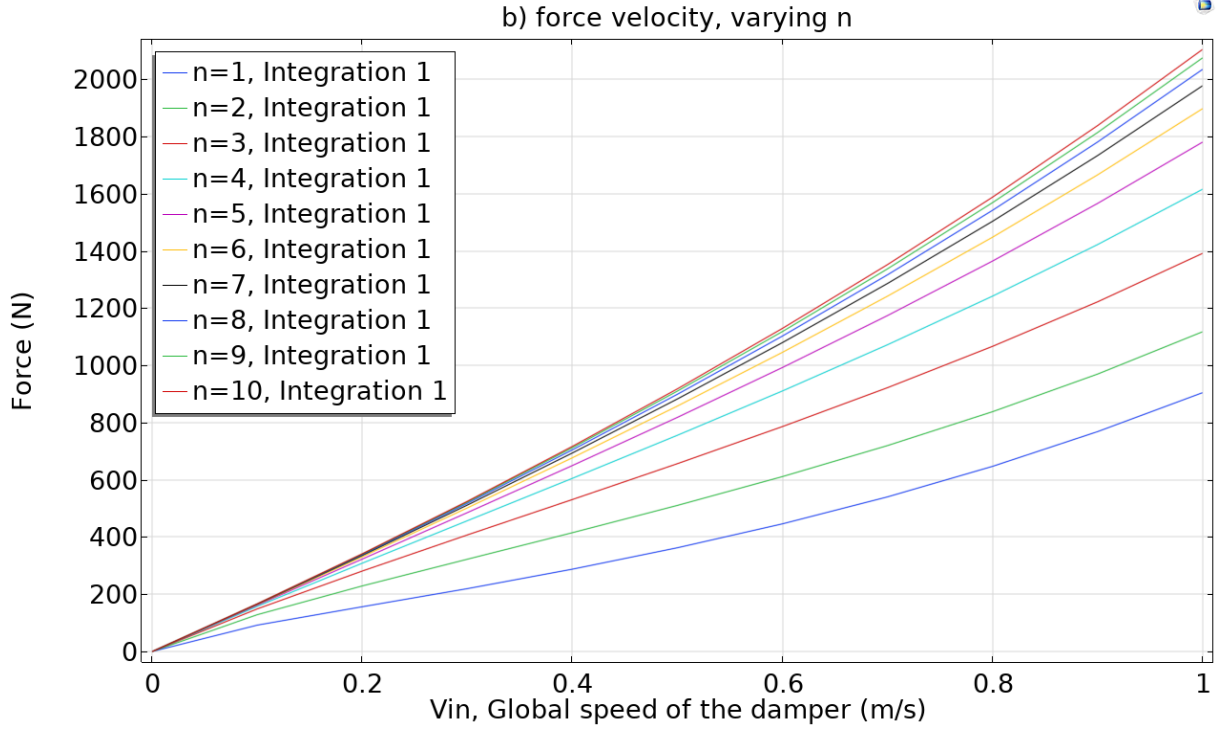


Figure 16: Force-velocity plot when number of shimstacks was changed from 1-10.

Compared to case (a) the amount of shims now affects the deflection of the shimstack and therefore increases the force with increasing n .

These plots were fitted to $F = C|v_{in}/v_0|^m$ with resulting values in Table 3 together with equivalent results for case (a) and (c).

The radius of the lower fixture RI (see Fig. 1) is changed from 9 to 14 in Figure 17. The plot shows that when RI is 9-13 mm, the difference in force-velocity plot is negligible, however at $RI = 14$ mm the gap becomes small enough that it impacts the force-velocity curve. An interesting observation is the fact that the other RI 's are not in a particular order, however these lines (plots) are so close together that the difference may be ignored.

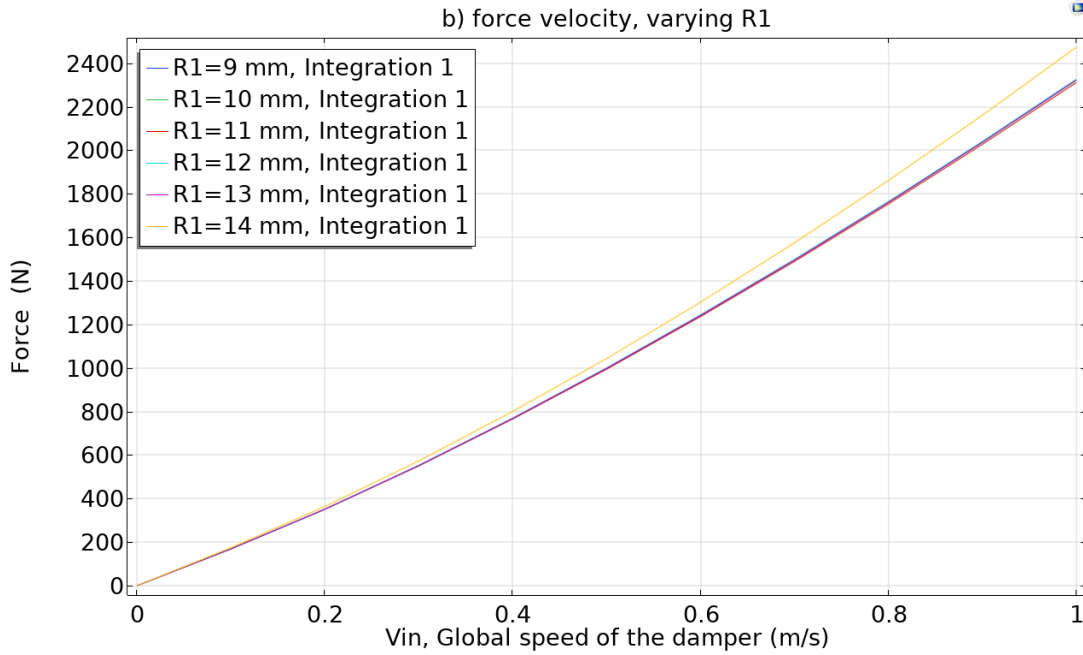
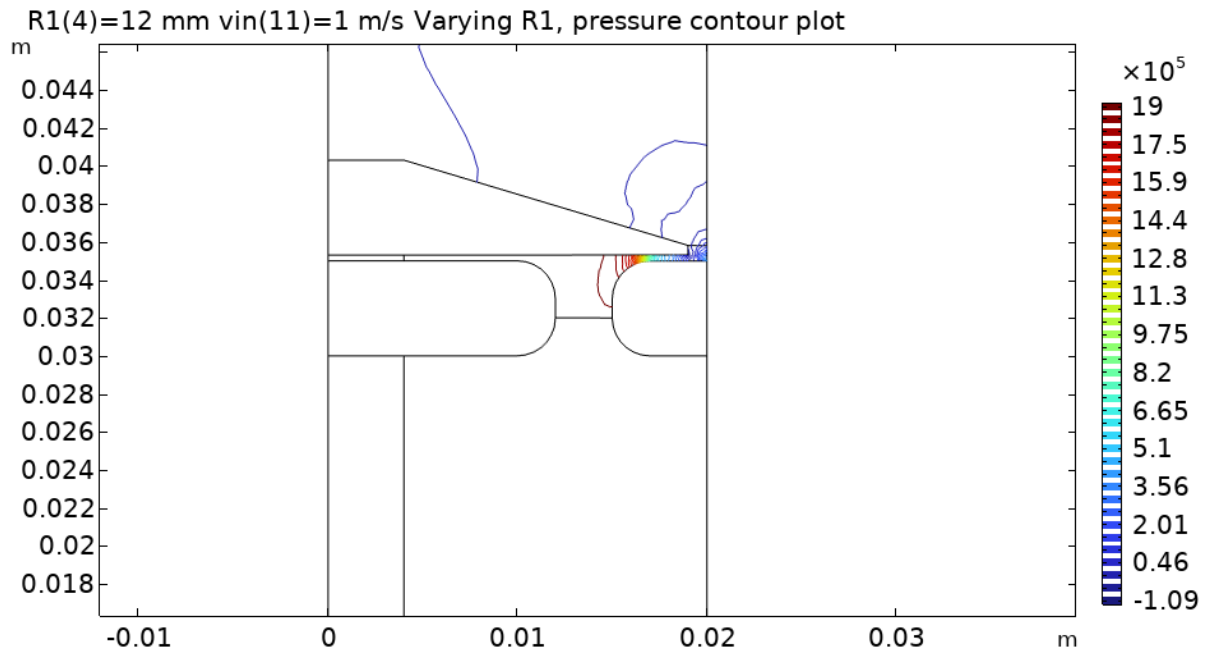


Figure 17: Force velocity plot when $R1$ is swept

When $R1 = 14$ mm the valve enters a critical flow region, meaning that flow is further constricted, which changes the system from a wide-gap flow regime to a thin-gap throttling regime. As is seen in Figure 18, a pressure gradient is also created near the part affected by $R1$, that is not observed when $R1$ is lower than 13 mm.



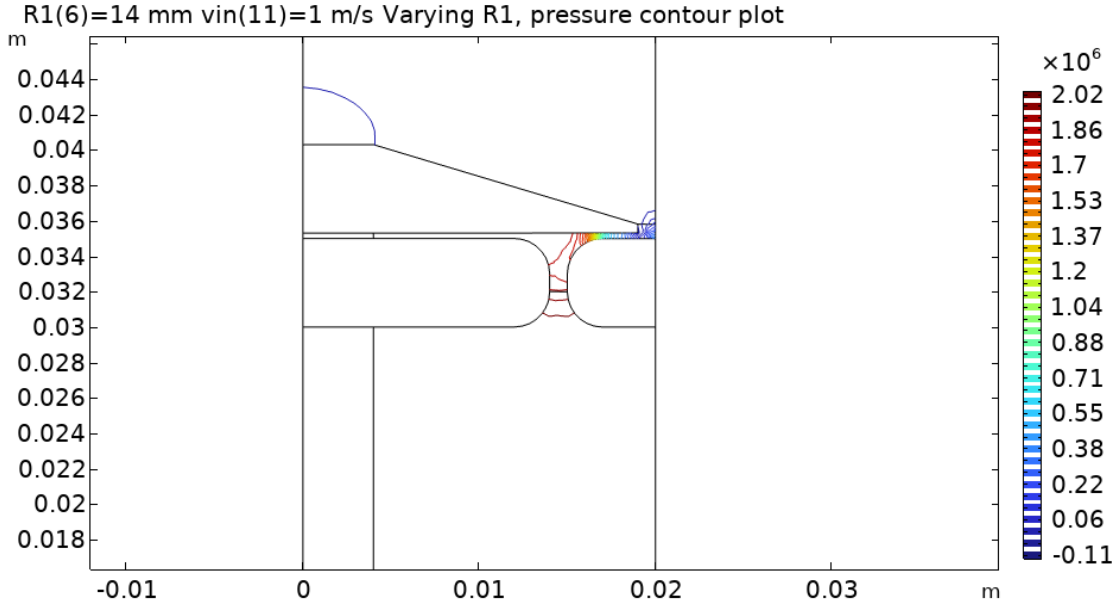


Figure 18: 3 plots showing the contour of pressure when $R1 = 12$ and 14 mm

The stress state at the shimstacks were lastly investigated and one can observe that the maximum stress occurs at the corner to the shimstacks at a value of 215 MPa, see Figure 19. The maximum stress at the region is reasonable considering the sharp corner.

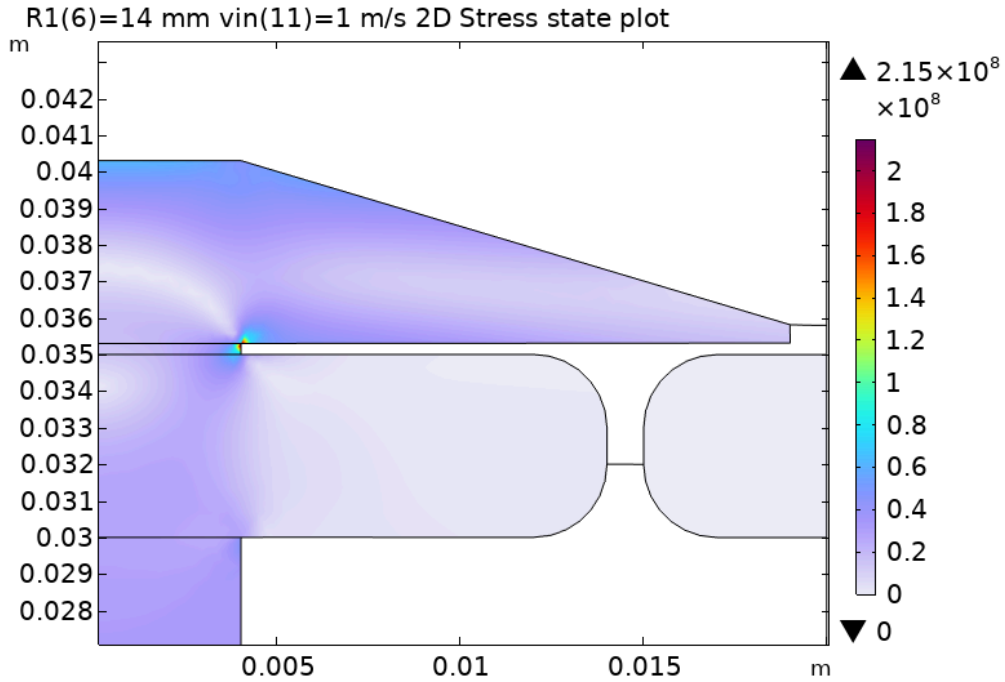


Figure 19: Stress state in the shimstack

4.3. Results for case c

Firstly a convergence study was done with varying *numMeshLayers* which affects the mesh according to chapter 3. Figure 20 shows the convergence for maximum velocity which is the

most critical case. After $numMeshLayers=3$ the graph changes around 4% which is sufficient, thus this value was used for the results. Convergence plots for all velocities are shown in Appendix [B].

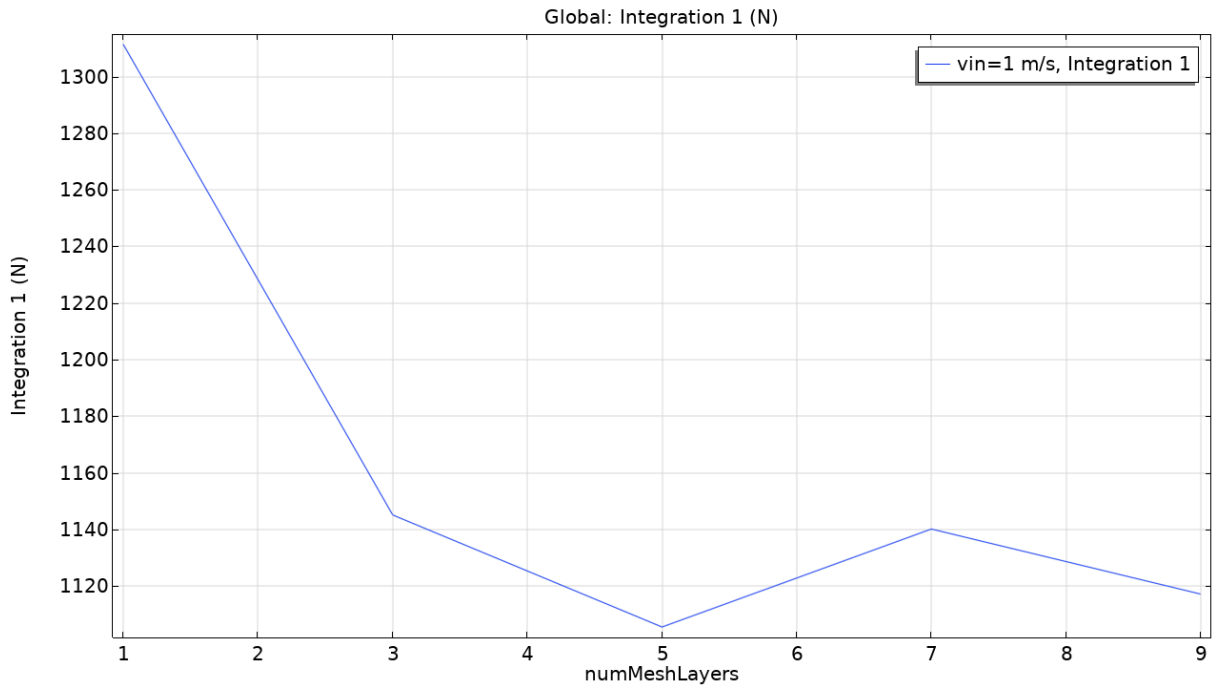


Figure 20: Mesh convergence plot for case (c)

The object was to investigate the effect on the force from different numbers of shims. With low n the shims deflect a lot which affects the mesh and result in convergence problems. For $n=1$ a solution was not found and for $n=2$ a solution was found up to $v_{in}=0.99875$ m/s where convergence problems occurred. The results from $n=2$ were plotted with the other cases using Matlab, see Figure 21.

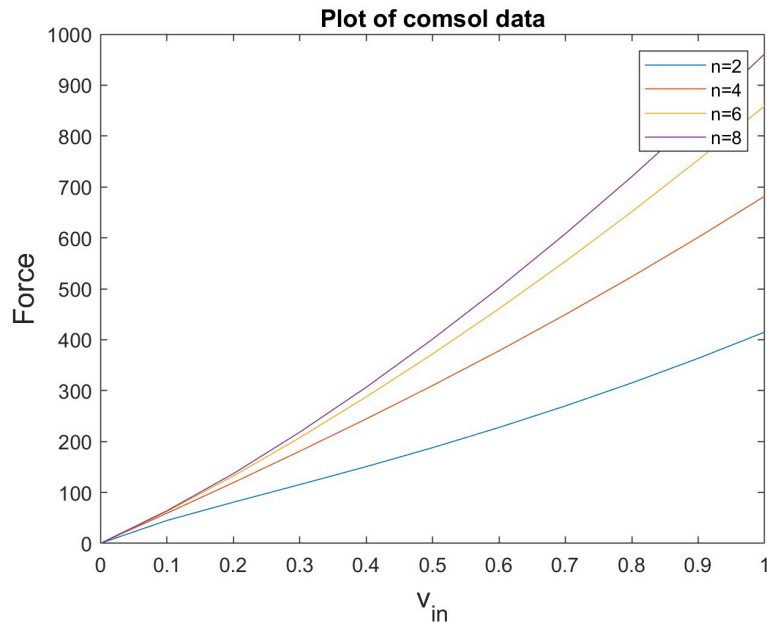


Figure 21: Force velocity plot for different number of shims.

The plots was fitted to $F = C|v_{in}/v_0|^m$ in the same manner as for the previous cases with results in Table 4.

Table 4: Table of m and C with varying number of shims for case (a), (b) and (c).

n [-]	m (a) [-]	C (a) [N]	m (b) [-]	C (b) [N]	m (c) [-]	C (c) [N]
2	1.278	2581	1.044	1080	1.074	405.5
4	1.278	2579	1.057	1589	1.109	674.9
6	1.278	2579	1.113	1872	1.187	852.8
8	1.278	2578	1.146	2008	1.240	954.3

It is noticeable that C is around half for case (c) compared to case (b). A large change in behaviour is expected since the geometry is different. A smaller C indicates lower force, this means that the assumptions made in case (a) and (b) overestimates the force. This is to be expected, the resistance is mainly due to viscous effects under the shimstack. While decreasing the flowpath, the viscous resistance in the flowpath increases which would increase the force, this will however weaken the shimstack since the pressure is more local. The shimstack will therefore deflect more over the flowpath, decreasing the viscous effects in this region. The oil has an easy flowpath under the left side of the shimstack in Figure (22)

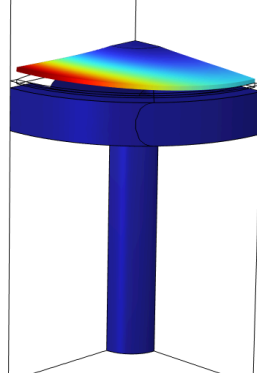


Figure 22: Displacement in case c with $n=4$, maximum displacement $13 \cdot 10^{-5} m$

Table 4 also shows that m is approximately the same for both (b) and (c), the assumption for axisymmetry in case (b) is thereby sufficient to determine the behaviour of the force velocity curve.

5. Conclusions

5.1: Case (a)

- The analytical force-velocity calculation matches COMSOL closely, confirming that continuity gives a reliable estimate for this laminar case.
- Mesh convergence is reached at around 8 boundary layers.
- Reducing $h/2a$ increases flow restriction which leads to higher force. Increasing $h/2a$ has the opposite effect.
- Curve-fitting yields $m \approx 1.2 - 1.3$, indicating that both viscosity and inertia influence the pressure drop (neither purely linear nor quadratic with velocity).
- Changing the number of rigid shims has little influence because the dominant restriction is geometric, not structural.
- When inertia is removed, the force-velocity curves become nearly linear and significantly smaller, showing that inertial effects are important in these narrow, high-velocity regions ($Re > 1$).

5.2: Case (b)

- Allowing the shimstack to deform reduces both C (force level) and m (velocity exponent), because deformation increases the effective flow area and lowers hydraulic resistance.
- Increasing the number of shims raises the force, since more shims reduce deformation and maintain a smaller flow gap.
- Sweeping RI shows almost no effect for $RI = 9-13$ mm, but at $RI = 14$ mm the gap becomes small enough to create a thin-gap throttling regime, sharply increasing force.
- The highest stresses occur at the sharp corner of the shimstack (≈ 215 MPa), which is expected due to geometric stress concentration.

5.3: Case (c)

- The physical geometry decreases the force compared to earlier cases.
- The m is similar to case (b) validating the shape of the force velocity curve.

6. AI

Generative Artificial intelligence, specifically ChatGPT, was used to answer quick theory explanations before being checked through more reliable sources. It was used to generate conclusions and an abstract. Text generated by ChatGPT was checked before being pasted into the document.

7. References

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8. Appendix

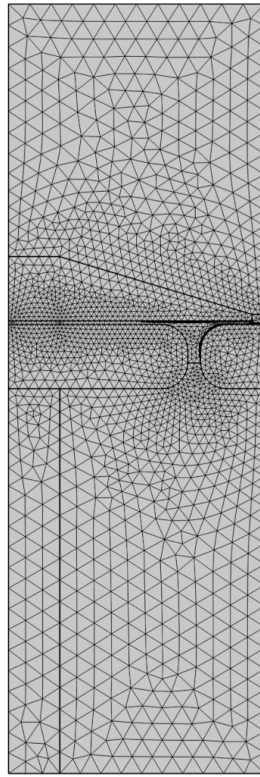


Figure A: Mesh used for case (b)

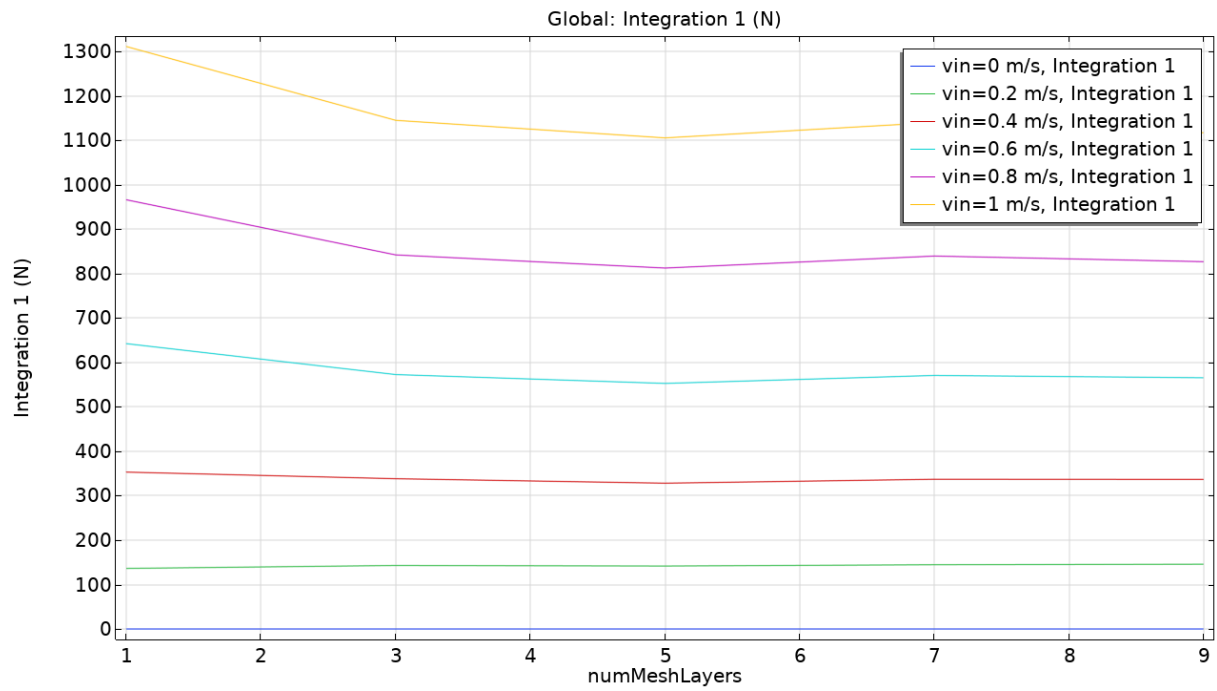


Figure B: Convergence for different velocities